

Immunodeficiency 67 (IRAK-4 Deficiency): Comprehensive Disease Characterization

Disease: Immunodeficiency 67 (IMD67) — Autosomal Recessive Human IRAK-4 Deficiency
MONDO ID: MONDO:0011888 | **OMIM:** #607676 | **Orphanet:** 70592 **Category:** Mendelian (autosomal recessive inborn error of innate immunity) **Causal gene:** *IRAK4* (12q12; HGNC:17967; NCBI Gene 51135; Ensembl ENSG00000198001; UniProt Q9NWZ3)

Summary

Immunodeficiency 67 is autosomal recessive human IRAK-4 deficiency, a Mendelian inborn error of innate immunity caused by biallelic loss-of-function mutations in *IRAK4*, the gene encoding interleukin-1 receptor-associated kinase 4. IRAK-4 is a Ser/Thr kinase that sits at the heart of the MyD88-dependent "Myddosome" signaling platform, transducing signals from nearly all Toll-like receptors (all except TLR3) and every member of the interleukin-1 receptor family. Complete loss of IRAK-4 function abolishes activation of NF- κ B and MAPK downstream of these Toll/IL-1 receptor (TIR)-domain receptors, so affected cells fail to mount inflammatory cytokine responses to purified TLR and IL-1R agonists.

Paradoxically, despite the breadth of receptors that depend on IRAK-4, the clinical phenotype is **narrow**: patients are selectively susceptible to a small set of pyogenic (pus-forming) bacteria, dominated by *Streptococcus pneumoniae*, with *Staphylococcus aureus* and *Pseudomonas aeruginosa* next in frequency. In the largest cohort (48 patients), invasive pneumococcal disease occurred in 68% of patients, the first invasive infection struck before age 2 in 88%, and mortality was high in early childhood (24 deaths). A clinical hallmark is **blunted or delayed inflammation** — patients often have no fever and weak acute-phase responses despite overwhelming invasive infection, which dangerously masks severity. Crucially, susceptibility and mortality **improve markedly with age**, presumably as adaptive immunity and anti-pneumococcal antibody develop.

Management is entirely **preventive and supportive**: continuous antibiotic prophylaxis, pneumococcal conjugate/polysaccharide vaccination, immunoglobulin replacement, caregiver vigilance, and immediate empiric broad-spectrum antibiotics at the first sign of infection. No

curative pharmacotherapy exists; hematopoietic stem cell transplantation is generally not indicated because prognosis improves with age, and there is no gene or cell therapy in clinical use. Breakthrough invasive disease can still be fatal even in fully vaccinated, prophylaxed patients. An emerging extension of the phenotype is heightened susceptibility to severe COVID-19, attributed to impaired TLR7-dependent type I interferon production.

Key Findings

1. Disease identity and genetic basis (F001, F010, F014)

Immunodeficiency 67 is the OMIM designation (#607676) for autosomal recessive human IRAK-4 deficiency. It is caused by **biallelic loss-of-function mutations in IRAK4** (interleukin-1 receptor-associated kinase 4; gene OMIM 606883; HGNC:17967; NCBI Gene 51135; Ensembl ENSG00000198001; UniProt Q9NWZ3), located at chromosome **12q12**. IRAK4 is essential for signaling downstream of most Toll-like receptors (all except TLR3) and all IL-1 receptor family members via the MyD88-dependent Myddosome complex, culminating in activation of NF-κB and MAPK. As a review states, "*interleukin-1 receptor-associated kinase 4 (IRAK4) plays a vital role in the TLR signaling cascade*" and "*Human IRAK4 deficiency is an autosomal recessive inborn error of immunity that classically presents with blunted or delayed inflammatory response to infection and susceptibility to a narrow spectrum of pyogenic bacteria*" [PMID: 32532880](#).

Complete identifier set (from MONDO:0011888): OMIM:607676; Orphanet:70592; MeSH C563662 & C564352; UMLS C1843256; MedGen 375137; GARD 0010311; NANDO:1200361/2200762. **Synonyms/alternative names:** IRAK-4 deficiency; IRAK4 deficiency; IRAK4D; immunodeficiency due to interleukin-1 receptor-associated kinase-4 deficiency; and the phenotype-based synonym "**recurrent isolated invasive pneumococcal disease, type 1 (IPD1)**". **ICD-11** maps to the category 4A00.1 (immunodeficiencies due to defects in innate immunity / Toll-like receptor signaling pathway defects); **ICD-10** maps to D84.8/D84.9 (other/unspecified immunodeficiency).

Information for this entry is derived predominantly from **aggregated disease-level resources** (OMIM, Orphanet, MONDO, HPO) and from published case series and case reports rather than individual EHR data.

2. Molecular mechanism: total loss of TIR-receptor signaling (F003, F008)

In the founding report (Picard et al., *Science* 2003), blood and fibroblasts from IRAK-4-deficient children *"did not activate nuclear factor kappaB and mitogen-activated protein kinase (MAPK) and failed to induce downstream cytokines in response to any of the known ligands of TIR-bearing receptors"* PMID: 12637671. Because the children were otherwise healthy except for pyogenic bacterial infections, the authors concluded that *"the TIR-IRAK signaling pathway is crucial for protective immunity against specific bacteria but is redundant against most other microorganisms"* PMID: 12637671. This defines both the mechanism (a complete block of TIR-receptor → NF-κB/MAPK signaling) and the reason for the surprisingly narrow infection susceptibility (redundancy against viruses, fungi, parasites, and most bacteria).

Protein architecture: IRAK4 (UniProt Q9NWZ3, 460 aa) has an **N-terminal death domain** that mediates recruitment to the MyD88-nucleated Myddosome, and a **C-terminal Ser/Thr kinase domain**. *"Upon ligand binding and via its N-terminal death domain, IRAK4 is recruited to an oligomeric receptor that is proximal to the Myddosome signaling complex, inducing IRAK4 kinase domain dimerization, autophosphorylation, and activation"* PMID: 30679311. Pathogenic variants span nonsense, frameshift, and splice-site alleles (LoF; the most common class — e.g., p.Gln293*/c.877C>T; splice c.942-1G>A) as well as missense variants in the kinase domain — all abrogating kinase activity or protein expression.

3. Clinical spectrum and outcome (F002, F011)

The largest cohort (Picard et al. 2010; 48 IRAK-4-deficient patients from 37 kindreds in 15 countries) established the natural history:

Feature	Value
Invasive pneumococcal disease	41/48 patients (68%); 52.2% of all invasive infections
Invasive <i>P. aeruginosa</i> / <i>S. aureus</i>	~16–17% each
First invasive infection before age 2	88.3%
First invasive infection in neonatal period	32.7%
Systemic inflammatory signs	Usually weak or delayed
Deaths	24
Recurrent invasive infection in survivors	36/50 (72%)

Key quotes: *"The leading threat was invasive pneumococcal disease, documented in 41 patients (68%) and causing 72 documented invasive infections (52.2%)"; "The first invasive infection occurred before the age of 2 years in 53 (88.3%) and in the neonatal period in 19 (32.7%) patients"; "Systemic signs of inflammation were usually weak or delayed"; "Multiple or recurrent invasive infections were observed in most survivors (n = 36/50, 72%)"; and "Clinical outcome was poor, with 24 deaths"* PMID: 21057262. Deaths clustered around the first invasive episode and around invasive pneumococcal disease, but invasive infections and deaths became rare after childhood — an **age-dependent improvement** in outcome.

Phenotype/HPO annotations: recurrent invasive bacterial infection (HP:0002719 Recurrent infections; HP:0002718 Recurrent bacterial infections), invasive pneumococcal disease with sepsis (HP:0100806) and meningitis (HP:0001287), cutaneous/skin abscesses mainly from *S. aureus* (HP:0100658; HP:0032243), upper respiratory tract infection (HP:0002788), septic arthritis (HP:0001369) and osteomyelitis (HP:0002754), and lymphadenitis (HP:0002840). Laboratory hallmarks: **absent/blunted fever** (HP:0001945 often ABSENT), weak acute-phase response with low or paradoxically normal CRP, and transient neutropenia during sepsis (HP:0001875). Immunoglobulins and lymphocyte subsets are typically normal. Severity is greatest in infancy and improves with age; the course is episodic/recurrent.

4. Population genetics: recessive, LoF-tolerant gene (F004, F013)

gnomAD constraint metrics for *IRAK4* confirm the gene is **not** haploinsufficient: pLI $\approx 2.5 \times 10^{-13}$ (essentially 0), observed/expected LoF ratio (oe_lof) = 0.85 (90% CI 0.67–1.09; LOEUF ~ 1.09), with 44 observed vs 51.8 expected LoF variants. This LoF tolerance is fully consistent with **autosomal recessive inheritance and asymptomatic heterozygous carriers**.

Inheritance is autosomal recessive with **complete penetrance** for the immunodeficiency but **highly variable expressivity** (age of first infection, organ site, survival). Estimated prevalence is **<1/1,000,000** (Orphanet:70592; ultra-rare); total published cases number in the low hundreds. Sex ratio $\sim 1:1$ (autosomal). Both consanguineous (homozygous) and non-consanguineous (compound heterozygous) kindreds are reported across many ethnicities — *"48 patients with IRAK-4 deficiency and 12 patients with MyD88 deficiency, from 37 kindreds in 15 countries"* PMID: 21057262. There is no genetic anticipation, mitochondrial inheritance, or recurrent large chromosomal abnormality. No single founder mutation dominates, though recurrent alleles (e.g., c.877C>T p.Gln293*) recur across unrelated families.

5. Extended phenotype: severe COVID-19 (F005)

An important recent extension of the phenotype: in a 22-patient series of AR MyD88 or IRAK-4 deficiency (mean age 10.9 yr) infected with SARS-CoV-2, 16/22 were hospitalized (6 moderate, 4 severe, 6 critical pneumonia, 1 death). *"The risk of invasive mechanical ventilation was also much greater than in age-matched controls from the general population (OR: 74.7, 95% CI: 26.8-207.8, $P < 0.001$)"* PMID: 36880831. The mechanism: *"The patients' susceptibility to SARS-CoV-2 can be attributed to impaired TLR7-dependent type I IFN production by pDCs"* PMID: 36880831. This links IRAK-4 to antiviral defense via the TLR7/plasmacytoid dendritic cell/type I interferon axis — a susceptibility not fully appreciated from the classic bacterial-only picture.

6. Human vs. mouse: a striking phenotypic contrast (F006)

Mouse *Myd88/Irak4* knockouts are broadly susceptible to many pathogens, whereas humans have a narrow phenotype. von Bernuth et al. note *"MyD88 deficiency in mice leads to susceptibility to a broad range of pathogens in experimental settings of infection,"* whereas human AR MyD88-deficient children suffered a narrow range of pyogenic bacterial infections and *"these patients were otherwise healthy, with normal resistance to other microbes. Their clinical status improved with age"* PMID: 18669862. The same contrast applies to *Irak4*-knockout mice versus human IRAK-4 deficiency. This has major implications for model-organism interpretation (see Section 15).

7. Diagnosis and management (F007, F009, F012)

Diagnosis rests on demonstrating **absent cellular responses to TLR/IL-1R (TIR) agonists** — no NF- κ B/MAPK activation and no cytokine (IL-6, TNF) induction in blood/fibroblasts PMID: 12637671 — confirmed by biallelic *IRAK4* mutation on sequencing (WES or gene panel). Rapid functional assays include flow-cytometric measurement of I κ B- α degradation after TLR stimulation (Frans et al. 2024, PMID: 37929815) and NF- κ B reporter assays in IRAK4-null HEK293T cells: *"We established a novel NF- κ B reporter assay using IRAK4-null HEK293T, which enabled the precise evaluation of IRAK4 mutations"* PMID: 33083971. A critical diagnostic clue is severe invasive infection with characteristically **low or delayed inflammatory signs** — patients often lack fever and have blunted acute-phase responses (though CRP can sometimes be elevated).

Anatomical structures affected (because IRAK4 is ubiquitously expressed, disease occurs wherever pyogenic bacteria invade): bloodstream (UBERON:0000178; sepsis/bacteremia), meninges/CNS (UBERON:0002360; meningitis, brain abscess), lungs (UBERON:0002048; pneumonia/empyema), skin and soft tissue (UBERON:0002097; abscesses, cellulitis), bone and

joints (UBERON:0002481; osteomyelitis, septic arthritis), lymph nodes (UBERON:0000029; lymphadenitis), and upper respiratory tract (UBERON:0001557). The Picard cohort noted infection "*with a high incidence of infections of the upper respiratory tract and the skin*" PMID: [21057262](#). Key cell types with defective TLR/IL-1R signaling: monocytes/macrophages (CL:0000576/CL:0000235), neutrophils (CL:0000775), dendritic cells including plasmacytoid DCs (CL:0000784), and non-hematopoietic cells such as fibroblasts and epithelial cells. Subcellular: the cytoplasmic Myddosome platform and endosomal TLR compartment (GO:0035325 Toll-like receptor binding; GO:0007249 canonical NF- κ B signal transduction).

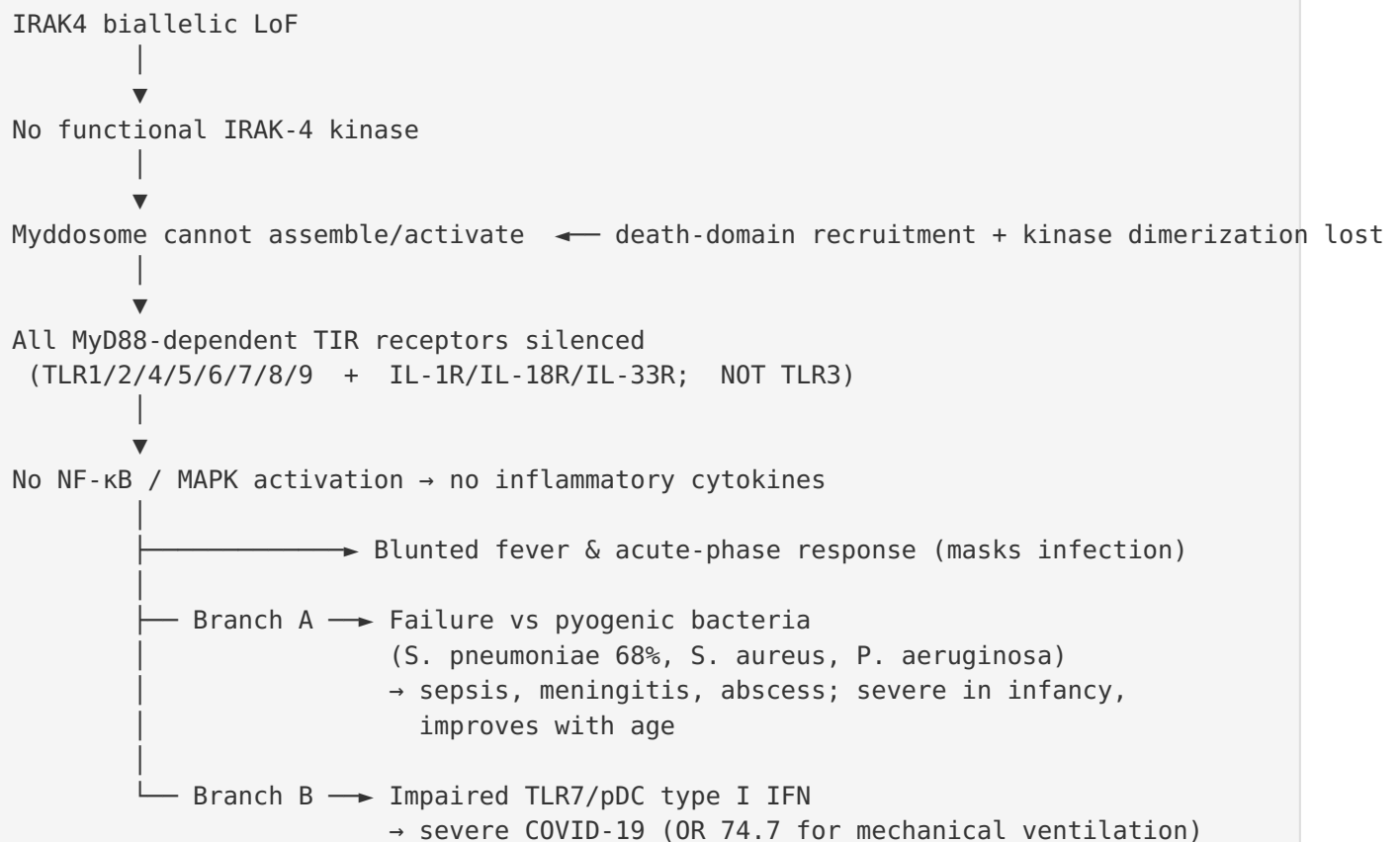
Management is preventive: continuous antimicrobial prophylaxis (e.g., penicillin V or trimethoprim-sulfamethoxazole), pneumococcal conjugate/polysaccharide vaccination (PCV13, PPSV23), often immunoglobulin replacement, caregiver vigilance, and immediate empiric broad-spectrum antibiotics at the first sign of infection. Yet breakthrough disease occurs: a vaccinated, prophylaxed girl "*was managed with antibiotic prophylaxis (sulfa/trimethoprim/ PenV, then - due to neutropenia - Cefprozil), pneumococcal vaccination (PCV-7, Pneumovax23, PCV-13) and vigilance*" but died of pneumococcal (serotype 6C) meningitis at age 7 PMID: [24596024](#). That report emphasizes that "*IRAK-4 deficiency causes IL-1R and TLR signaling failure, resulting in minimal clinical features despite invasive bacterial infection*" PMID: [24596024](#) — the central reason vigilance and empiric treatment are essential.

Mechanistic Model / Causal Chain

Ordered causal chain from mutation to clinical manifestation:

1. **Biallelic loss-of-function mutation in *IRAK4* (12q12) leads to** absent or non-functional IRAK-4 kinase protein.
2. Absence of functional IRAK-4 **results in** failure to assemble/activate the MyD88-nucleated **Myddosome** — IRAK-4 can no longer be recruited via its death domain, dimerize, autophosphorylate, and activate downstream IRAK1/IRAK2 PMID: [30679311](#).
3. Myddosome failure **abolishes** signal transduction from all TIR-domain receptors that use MyD88 — **all TLRs except TLR3** and **all IL-1R family receptors** (IL-1R, IL-18R, IL-33R).
4. This **results in** failure to activate **NF- κ B** and **MAPK**, so cells "*failed to induce downstream cytokines in response to any of the known ligands of TIR-bearing receptors*" PMID: [12637671](#).

5. Loss of pro-inflammatory cytokine induction (IL-6, TNF, IL-1 β) **leads to a blunted/absent acute-phase response and fever** — invasive infection proceeds with "minimal clinical features."
6. **Branch A (dominant — bacterial):** Impaired early innate sensing of pyogenic bacteria (especially encapsulated *S. pneumoniae*) **leads to** failure to contain infection at mucosal/entry sites → invasive bacteremia, meningitis, pneumonia, abscesses. This is worst in infancy, before protective anti-pneumococcal antibody develops, and **improves with age** as adaptive immunity matures.
7. **Branch B (viral, inferred/emerging):** Loss of **TLR7-dependent type I IFN production by plasmacytoid dendritic cells results in** susceptibility to severe SARS-CoV-2 pneumonia [PMID: 36880831](#).



Upstream vs downstream: the *IRAK4* mutation and Myddosome failure are the most upstream lesion; NF- κ B/MAPK silencing is the proximal molecular consequence; blunted inflammation and impaired bacterial/viral containment are the downstream clinical manifestations. **GO terms:** GO:0007249 (canonical NF- κ B signal transduction), GO:0002224 (Toll-like receptor signaling pathway), GO:0035325 (Toll-like receptor binding), GO:0070498

(interleukin-1-mediated signaling), GO:0045087 (innate immune response), GO:0032496 (response to lipopolysaccharide). **Relevant CHEBI:** lipopolysaccharide, lipopeptide, CpG oligodeoxynucleotide (TLR agonists whose signaling is lost).

Evidence Base

PMID	Title (abbreviated)	Evidence type	How it supports findings
12637671	<i>Pyogenic bacterial infections in humans with IRAK-4 deficiency</i>	Human clinical + in vitro	Founding report. Establishes the total TIR-signaling defect (no NF- κ B/MAPK, no cytokines) and the narrow-but-crucial role of the pathway.
21057262	<i>Clinical features and outcome of patients with IRAK-4 and MyD88 deficiency</i>	Human clinical (n=48)	Largest cohort. Natural history: 68% pneumococcal, onset <2 yr in 88%, 24 deaths, weak inflammation, age-dependent improvement.
32532880	<i>Clinical IRAK4 deficiency...</i>	Review + case	Defines disease as AR inborn error with blunted inflammation and narrow pyogenic susceptibility; IRAK4's vital TLR role.
36880831	<i>Humans with inherited MyD88 and IRAK-4 deficiencies predisposed to hypoxemic COVID-19</i>	Human clinical (n=22)	Extends phenotype to severe COVID-19 (OR 74.7 for ventilation) via impaired TLR7/pDC type I IFN.
18669862	<i>Pyogenic bacterial infections in humans with MyD88 deficiency</i>	Human clinical + comparative	Documents the human-vs-mouse contrast: broad murine susceptibility vs narrow human phenotype improving with age.
30679311	<i>Conformational flexibility and inhibitor binding to unphosphorylated IRAK4</i>	Structural	Death-domain recruitment + kinase-domain dimerization/ autophosphorylation mechanism; basis for how LoF variants disrupt function.
33083971	<i>IRAK4 Deficiency Presenting with Anti-NMDAR Encephalitis and HHV6 Reactivation</i>	In vitro assay + case	NF- κ B reporter assay in IRAK4-null HEK293T for precise variant evaluation (functional confirmation).
24596024	<i>Fatal pneumococcal meningitis in a 7-year-old... despite prophylaxis</i>	Human case	Documents standard prophylactic regimen and breakthrough fatal disease; blunted inflammation despite invasive infection.

PMID	Title (abbreviated)	Evidence type	How it supports findings
37929815	<i>Diagnosis of IRAK-4-deficiency by flow cytometric IκB-α degradation</i>	Diagnostic method	Rapid functional diagnostic assay.
42103176	<i>Compound het IRAK4 variants: bacterial infections, brain calcification, epilepsy</i>	Human case + functional	Expands variant/phenotype spectrum (frameshift c.123dupA + missense c.543T>G); persistently elevated CRP variant; reduced protein and impaired TLR signaling.
26472314	<i>IRAK-4 deficiency misdiagnosed as AD Hyper-IgE syndrome</i>	Human case	Illustrates diagnostic pitfalls (recurrent <i>S. aureus</i> skin infections, elevated IgE); homozygous c.877C>T p.Gln293*.
38838930	<i>Two novel compound het LoF mutations, fatal P. aeruginosa sepsis</i>	Human case + functional	Splice/frameshift variants (c.942-1G>A) confirmed by minigene assay; fatal neonatal <i>Pseudomonas</i> sepsis.

Additional supporting case reports document the anatomical/clinical breadth: transcranial abscess management [PMID: 25569407](#), anti-NMDAR encephalitis with HHV6 reactivation [PMID: 33083971](#), CNS pseudomonal vasculopathy [PMID: 39846126](#), *Salmonella* osteomyelitis [PMID: 38857180](#), adult fatal meningitis [PMID: 37103729](#), and delayed adult presentation [PMID: 38758474](#). The systems-immunology study [PMID: 25344726](#) showed that responses to purified agonists are globally abolished but "*variable residual responses were present following exposure to whole pathogens*," identifying "*a narrow repertoire of transcriptional programs affected*" — the molecular correlate of the narrow clinical phenotype.

Section-by-Section Data Compilation

Etiology

Purely **genetic**: biallelic (homozygous or compound heterozygous) LoF mutations in *IRAK4*. No environmental cause. **Risk factor** for expression: consanguinity (increases homozygosity). Infectious exposure to encapsulated pyogenic bacteria is the trigger for clinical events but not a cause of the underlying condition. Heterozygous carriers are unaffected (gnomAD LoF tolerance). No known protective genetic modifiers are characterized; the strongest **protective factor** is increasing age (maturation of adaptive/antibody immunity). Gene-environment interaction: the genetic defect determines susceptibility, while environmental pathogen exposure and vaccination status determine timing/severity of clinical events.

Environmental Information

No toxins, radiation, or occupational exposures are implicated. The relevant **infectious agents** are the pathogens that exploit the immune defect: *Streptococcus pneumoniae* (dominant), *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Salmonella* (osteomyelitis case), and SARS-CoV-2 (severe COVID-19). These are downstream complications, not causes.

Temporal Development

Onset: congenital genetic defect; first invasive infection typically in infancy — before age 2 in 88.3%, neonatal in 32.7%. **Pattern**: episodic/recurrent invasive infections. **Progression**: highest morbidity and mortality in early childhood, with **age-dependent improvement**; invasive infections and deaths become rare after childhood. **Critical period**: the first few years of life (before protective anti-pneumococcal antibody develops) is the window of greatest vulnerability and the key window for prophylactic intervention.

Outcome / Prognosis

Poor in infancy — 24 deaths in the 48-patient cohort, most during the first invasive episode or from invasive pneumococcal disease. Survivors have recurrent invasive infection (72%) but progressively improving prognosis with age. No formal 5-/10-year survival figures beyond cohort mortality; life expectancy approaches normal in patients who survive childhood with good prophylaxis. **Prognostic factors**: age (younger = worse), pathogen (pneumococcal worst), and access to prophylaxis/vaccination/vigilant empiric therapy.

Treatment (NCIT-relevant interventions)

- **Antibiotic prophylaxis** — penicillin V, trimethoprim-sulfamethoxazole (NCIT: Antibiotic Therapy).
- **Immunoglobulin replacement therapy** (IVIG/SCIG; NCIT: Intravenous Immunoglobulin Therapy).
- **Pneumococcal vaccination** — PCV13 conjugate, PPSV23 polysaccharide (NCIT: Vaccine Therapy); antibody responses can be satisfactory but short-lived, requiring boosting.
- **Immediate empiric broad-spectrum antibiotics** at first sign of infection (essential given blunted signs).
- **Surgical drainage** of abscesses (e.g., transcranial abscess, [PMID: 25569407](#)).
- **No curative pharmacotherapy; HSCT generally not indicated** (prognosis improves with age); no gene/cell therapy in clinical use.

Prevention

Primary: carrier screening/genetic counseling in affected families; vaccination; prophylactic antibiotics. **Secondary:** cascade genetic testing of siblings, early diagnosis, vigilant monitoring. **Tertiary:** aggressive early treatment of infections to prevent complications. **Counseling:** autosomal recessive — 25% recurrence risk for siblings of a proband; asymptomatic carriers; prenatal/preimplantation testing feasible where the familial variants are known.

Other Species / Model Organisms

- **Taxonomy:** disease as such is human (NCBI Taxon 9606). Orthologs: *Irak4* in mouse (NCBI Taxon 10090; Gene 266632) and rat.
- **Model organisms:** *Irak4* and *Myd88* knockout **mice** are the principal models. **Critical limitation:** murine knockouts show **broad** pathogen susceptibility, poorly recapitulating the **narrow** human phenotype [PMID: 18669862](#) — human disease is far more restricted and improves with age. **In vitro models:** patient fibroblasts/blood cells (loss of NF- κ B/MAPK response) and IRAK4-null HEK293T reporter lines for variant functional testing [PMID: 33083971](#). This species discordance is itself a key scientific finding: it demonstrates greater redundancy of the TIR-IRAK4 pathway in humans than in mice.

Limitations and Knowledge Gaps

1. **Rarity limits epidemiology.** With a prevalence <1/1,000,000 and only low-hundreds of published cases, prevalence/incidence estimates are imprecise and demographic/geographic patterns are anecdotal. No population-based registries provide robust survival curves.
 2. **The narrow-phenotype paradox is incompletely explained.** Why loss of a pathway serving nearly all TLRs and IL-1R members produces such restricted susceptibility (chiefly *S. pneumoniae*) remains only partly understood; residual whole-pathogen responses [PMID: 25344726](#) and redundancy with TLR3/TRIF and other pathways are implicated but not fully mapped.
 3. **Age-dependent improvement mechanism is inferred, not proven.** The favorable evolution with age is attributed to maturing adaptive immunity/antibody, but the precise immunological switch has not been directly demonstrated.
 4. **Viral susceptibility scope is still emerging.** The COVID-19 association [PMID: 36880831](#) and scattered viral reports (HHV6 reactivation, [PMID: 33083971](#)) suggest the antiviral phenotype (TLR7/pDC/type I IFN) is broader than the classic bacterial picture, but its full extent is undefined.
 5. **No curative therapy or genotype-phenotype map.** There is no gene/cell therapy, and correlation between specific *IRAK4* alleles and severity is not established (variable expressivity noted but not systematically modeled).
 6. **Model-organism discordance** limits translational research — mouse models over-predict susceptibility breadth.
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Proposed Follow-up Experiments / Actions

1. **Build a systematic *IRAK4* variant–function database.** Use the *IRAK4*-null HEK293T NF- κ B reporter [PMID: 33083971](#) and I κ B- α degradation flow assay [PMID: 37929815](#) to functionally classify all reported alleles, establishing a genotype-phenotype/severity map and ACMG-grade functional evidence.
2. **Define the antiviral phenotype prospectively.** Given the COVID-19 finding, systematically assess TLR7/pDC type I IFN responses and viral infection history across the patient

population to determine whether antiviral prophylaxis/vaccination guidance should change.

3. **Dissect the narrow-phenotype paradox.** Use single-cell transcriptomics of patient immune cells challenged with whole *S. pneumoniae* vs. other pathogens to identify the MyD88-independent (e.g., TLR3/TRIF, inflammasome, complement) compensatory circuits that spare non-pneumococcal defense.
4. **Characterize the age-dependent improvement.** Longitudinal profiling of anti-pneumococcal antibody repertoire and memory B/T cell maturation in surviving patients to test whether adaptive antibody accrual explains the falling infection rate — informing when prophylaxis may be safely de-escalated.
5. **Improve model systems.** Develop humanized or organoid/iPSC-derived myeloid models that better recapitulate the human narrow phenotype than mouse knockouts, for preclinical testing of interventions.
6. **Newborn/cascade screening feasibility.** Evaluate whether a TLR-agonist functional screen or targeted sequencing of *IRAK4/MYD88* is cost-effective in high-consanguinity populations or in siblings of probands, given the very high early-childhood mortality that early prophylaxis could prevent.

Conclusion

Immunodeficiency 67 is autosomal recessive human IRAK-4 deficiency: biallelic loss-of-function *IRAK4* mutations abolish MyD88/Myddosome-dependent TLR (all except TLR3) and IL-1-receptor-family signaling to NF- κ B and MAPK, producing a narrow but life-threatening susceptibility to pyogenic bacteria — above all invasive pneumococcal disease — with characteristically blunted inflammation, early-childhood onset and high early mortality that improves markedly with age, plus an emerging risk of severe viral disease (COVID-19). Management is entirely preventive (antibiotic prophylaxis, pneumococcal/conjugate vaccination, immunoglobulin, and immediate empiric antibiotics), as no curative therapy exists.